

Delft University of Technology

Faculty of Civil Engineering and Geosciences

AESM5230C · Coastal Remote Sensing

Estimating 3D Vegetation Dynamics in Point Clouds

Project report

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1 Introduction

Vegetation on young coastal dunes, such as those of the Zandmotor, traps wind-blown sand and holds it in place, so where plants grow, dunes grow. To understand and manage this, we need to know where the plants are, how large they are, how much biomass they hold, and how this changes through the year and from year to year. Satellite indices such as NDVI show greenness, but not the 3D size of plants. UAV LiDAR measures the surface in 3D with a point every few centimetres and can be repeated, so it can follow single plants through time.

In this report we ask four things: (1) What do the LiDAR point clouds look like, and how can we separate vegetation from sand? (2) How big are single plants, and how much biomass do they hold? (3) How reliable are the results? (4) How do vegetation and the ground change over the 2.7 years of data? We describe the data and the field data (Section 3), the methodology and how we validate it (Section 4), the results (Section 5), and we discuss and conclude (Sections 6 and 7).

2 Literature Review

Several algorithms have been developed to separate ground and non-ground object in 3D point cloud. These algorithms include Cloth Simulation Filtering (CSF), Progressive Morphological Filter (PMF), and Adaptive triangulated Irregular Network (ATIN) [1–3]

Several studies have conducted comparative studies between the three algorithms. ATIN is found to be most effective for high-density data in highly variant topographic landscape, yet struggles in flat or urban areas. Yet this limitation on flat terrain is challenged by P. P. Sulung and B. Soeksmantono [4]. Additionally, CSF and PMF are more flexible, yet performs less well in landscape with high terrain complexity, with CSF also performing well in urban settings [4–7].

Methods of biomass estimation, has developed throughout the years, however, its effectiveness to achieve highly accurate estimation depends on our knowledge in type of dataset available and examined, vegetation type, or landscape type. According to R. Bolanos *et al.* [8], biomass estimation studies often rely on allometric equations, volume-based reconstructions, or parametric regression models. Its method however can be reconstructed through few stages, including; exploration of vegetation characteristic, point-cloud reconstruction, biomass estimation, and lastly statistical analysis for validation purposes. To assess dataset attribute, few metrics are conducted to assess its characteristics. Previous studies, extracted structural metrics such as Diameter at Breast Height (DBH),tree height, volume and vegetation characteristic (vegetation type, tree, shrub species, or grassland) [8]. Topographic elements, such as DSM, DTM, slope, and CHM are extracted to estimate vegetation volume or understand its influence in biomass growth [9], [10].

Once vegetation or landscape attributes are obtained, researcher can choose different method of estimation. Common methods can be grouped as allometric, volumetric reconstruction (e.g.Convex Hull, Voxelisation, TIN) and machine learning methods (e.g.Random Forest/RF or Artificial Neural Network/ANN) .

Research conducted across various studies also found smaller differences between observed and fitted values (higher R^2) on allometry and volume-based approaches, followed by regression-based models and lastly, Machine Learning models (RF and ANN) in studies estimating biomass. The combination of allometry and volume-based approach is also often done. Within

allometry and volume-based approaches, voxelisation, convex hull, TIN, or alpha shape are most often mentioned in terms of 3D reconstruction. Moreover, vegetation type affect the method of volumetric reconstruction. For example, tree-based studies often opt for TIN, alpha shape or use of height metric. Whereas DSM, DTM, or volumetric computation using structural variation are often used for grassland, non-woody vegetation types [9, 11].

Voxel-based methods demonstrated moderate to high accuracy across vegetation types, despite being sensitive to its paramter [12–19]. The second approach to follow is a Convex-Hull-based approached, achieving a nearly unbiased biomass estimates and is close to outperforming voxelisation [20, 21]. The method however presents limitation of oversimplification of biomass estimation in dense or highly porous vegetation structures [8]. It is good to however acknowledge that the accuracy of biomass estimation and volumetric reconstruction depends heavily on the type of vegetation analyzed. This includes quality of the point cloud, vegetation structure, biomass density value, and the objective of the study itself [8].

3 Site Description

The Zandmotor is a 21.5 million m^3 sand nourishment on the Dutch coast between Hoek van Holland and Scheveningen, built in 2011 so that waves, wind and currents spread the sand along the coast by themselves. Over time the sand has formed beaches, a lagoon and young dunes with pioneer vegetation. Our study area is a 220 × 230 m box (x 72290 to 72510, y 451410 to 451640 in RD New, about 52.046° N, 4.183° E, (Figure 1). It holds a flat bare plain in the middle and two belts of low, hummocky dunes with scattered plants on the west side and the north-east side. All results in this report are limited to this box.

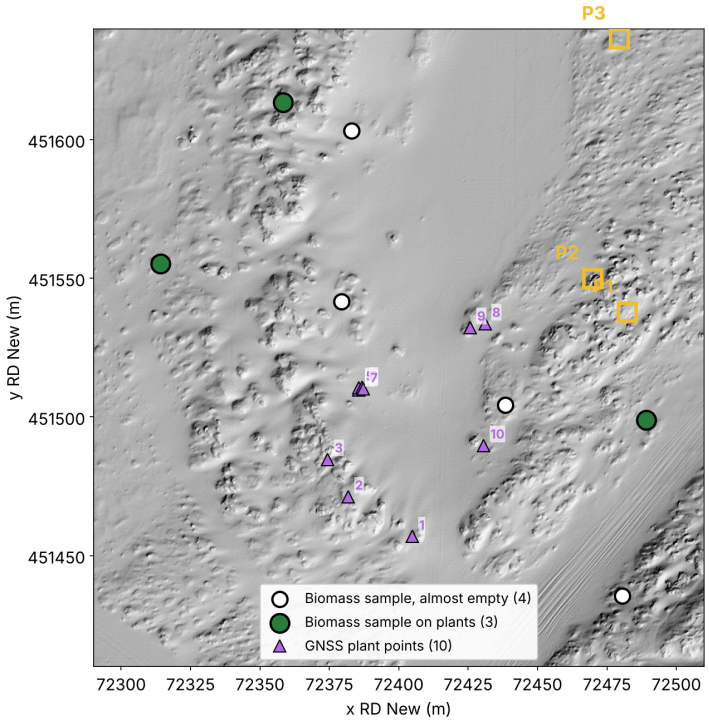


Figure 1: The study area (RD New coordinates; hillshade of the terrain from the LiDAR) with the seven biomass samples, the ten GNSS plant points and the three example plants P1 – P3 of Section 5.2.

Why this tile. It is the area around the seven biomass samples of Sander Vos, our only field data on biomass, and it holds ten of the sixteen GNSS plant points we recorded in the field. It is covered by all eight flights. Field points outside the box (six GNSS points) are not used, so that everything is tested on the same ground that we map.

3.1 UAV LiDAR flights

We have eight flights (Feb, May, Jun, Jul 2024; Mar, May 2025; Feb, Oct 2026), delivered as LAZ files in RD New (EPSG:28992) with heights in m NAP. Each point has x, y, z, RGB colour and intensity. All points are single returns, so a pulse gives only one point and we cannot see through the canopy. Table 1 shows that the flights of 2024 are 1.4 - 2.3 times denser than the flights of 2025 - 2026. This matters for comparisons in time, and we deal with it in Section 4.1.

Flight	Points in the study area	Density (pts/m²)	Point spacing (cm)	z range (m NAP)
epoch	points	density (pts/m2)	spacing (cm)	z range (m)
Feb 2024	43296293	855.7	3.4	2.16-9.16
May 2024	65886474	1302.1	2.8	2.07-12.64
Jun 2024	65496455	1294.4	2.8	2.02-18.96
Jul 2024	52696020	1041.4	3.1	1.75-18.33
Mar 2025	30173644	596.3	4.1	1.95-12.59
May 2025	29044384	574.0	4.2	1.96-12.62
Feb 2026	30206862	597.0	4.1	1.67-12.60
Oct 2026	32209292	636.5	4.0	1.65-12.52

Table 1: The eight flights, original data inside the study area.

3.2 Field data

- **Biomass samples.** Seven samples taken by Sander Vos on 27 March 2025 (the day of the March 2025 flight), given as dry biomass per m^3 of sampled volume. Three samples are on plants (1.50, 1.64 and 8.65 kg/m^3). Four are almost empty (0.02 - 0.09 kg/m^3), so only the three can be used to convert volume to biomass.
- **GNSS plant points.** Sixteen points recorded with RTK GNSS in the field on 7 October 2026 (the day of the Oct 2026 flight), 10 of them inside the box. The antenna stands on the ground at the base of a plant, so each point gives a ground height at a known place. The heights are ellipsoidal heights, not NAP (see Section 4.5). No species were recorded.

4 Methodology

The whole workflow is written in Python (`numpy`, `scipy`, `laspy`, `cloth-simulation-filter`, `scikit-learn`) in one helper file called from three notebooks (A, B, C, appended). The table lists the steps in the order they are done, the next sections explain the choices.

Step	Description and Parameters
1 Merge, crop, thin	clips per epoch merged and cropped to the study box; thinned to 250 pts/m^2
2 Outliers	points far below their neighbours are removed (they would pull the cloth down)
3 CSF	cloth resolution 0.5 m, rigidness 2, threshold 0.10 m → "ground" and "candidate" points
4 DTM	median ground height per 0.25 m cell, holes filled by interpolation; slope from the DTM
5 Height gate	height above DTM between 0.10 m (≈ 3 × the 3 cm noise of bare sand, slightly more on slopes) and 1.5 m
6 Shape gate	20 cm neighbourhood; vegetation if *not planar* (planarity $<$ 0.65 and (linearity $>$ 0.30 or sphericity $>$ 0.15)); sparse points ($<$ 8 neighbours) count as vegetation
7 Final DTM	candidates rejected only because they are too low go back to the ground; DTM rebuilt and gates 5–6 repeated. Flat candidates well above the ground (plant mat) are *not* added to the ground
8 Tussocks	DBSCAN on x,y of vegetation points, eps 0.20 m, min 8 points, ≥ 30 points per cluster, 0.10 – 1.5 m high, not slender (height / √footprint ≤ 1.5)
9 Volume, biomass	four volumes per tussock (column, convex hull, voxel 5 cm, alpha shape r = 0.15 m); $[B_{\text{est}} = V_{\text{column}}\varrho]$ with ϱ = 1.5 / 1.64 / 3.3 kg/m^3 (low / mid / high)
10 Change	cover and biomass per epoch on the area covered by all epochs; ground change = DTM difference with a 95 % level of detection from flat cells

Table 2: Workflow and parameters.

4.1 Point density and thinning

The flights have different densities (574 to 1302 pts/m^2, Table 1). The number of plants we detect depends on density (Section 5.4), so comparing a dense flight with a sparse one would show a "change" that is only a change in density. We therefore use two rules:

- **One flight on its own** (maps, plant examples, validation of one date, biomass of one date): all points are used, with no thinning, so that we do not underestimate small plants.
- **Comparing flights** (time series, change maps): every flight is randomly thinned to 570 pts/m^2, just below the sparsest flight (574 pts/m^2, May 2025), so that all flights have the same density.

4.2 Separating vegetation from sand

The vegetation in out area is low (10 cm to 1.3 m) and sparse, on dune slopes. A single height threshold cannot work on a slope. We therefore combine three things. First, CSF follows the terrain and gives a first ground and a set of candidate plant points. Second, a height gate removes candidates that are too close to the ground to be told apart from sand noise. Third, a shape gate keeps only candidates that are not flat. A flat candidate just above the ground

is more likely sand ripple than a plant. Where the ground under a plant is hidden, the ground model is interpolated from the sand around it. The result is checked in Sections 5.4 and 5.6.

4.3 Choice of parameters

The height gate (0.10 m) is three times the noise measured on bare sand (3 cm). The shape limits (planarity, linearity, sphericity) were chosen by looking at the three example plants and bare sand (Section 5.2). For CSF we tested what happens when each setting is changed (tab-variants) and kept the setting that gave no vegetation below 10 cm (which would be sand), kept the large plants together, and found vegetation at the three samples that have plants. The effect of the settings is part of the uncertainty we report.

5 Results

5.1 Point cloud characteristics

Table 1 gives the size of each flight. A flight has 30 - 66 million points in the box. Point spacing is 2.8 - 4.2 cm. Heights range from about 1.7 to 12 m NAP; the bare sand lies at 2 - 8 m NAP and the highest points (above 9 m) are outliers and objects, not terrain. Because all returns are single, there is no penetration of the canopy, and the LiDAR only sees the top of a plant and the sand between plants.

Effect of point density. To test how much density matters, we took the densest flight (May 2024), cut a 60 × 60 m area with much vegetation, thinned it to different densities and ran the full separation each time (Table 3).

Density (pts/m²)	**Vegetation points**	**Vegetation cover (m²)**
density (pts/m2)	vegetation points	vegetation cover (m2)
30	989	4.7
60	1644	7.9
125	2738	11.5
250	5382	17.5
369	8621	22.2

Table 3: Detected vegetation against point density in the same 60 × 60 m area. The area itself has 369 pts/m² that is the maximum we test.

Detection rises with density: cover grows from 4.7 m² at 30 pts/m² to 17.5 m² at 250 pts/m² and 22.2 m² at 369 pts/m², so even our densest data may still miss small plants. Low plants need several points inside the 10 cm height range, and enough neighbours for the shape test. This is the reason why we do not thin single flights, and why we compare flights at the same density.

5.2 Vegetation types and their features

We picked three plants from the detected vegetation of March 2025 by shape: the largest clump (Plant 1), the most elongated (Plant 2) and the most compact (Plant 3). Their positions are in Figure 1 and their shape is in Figure 2 and Table 4.

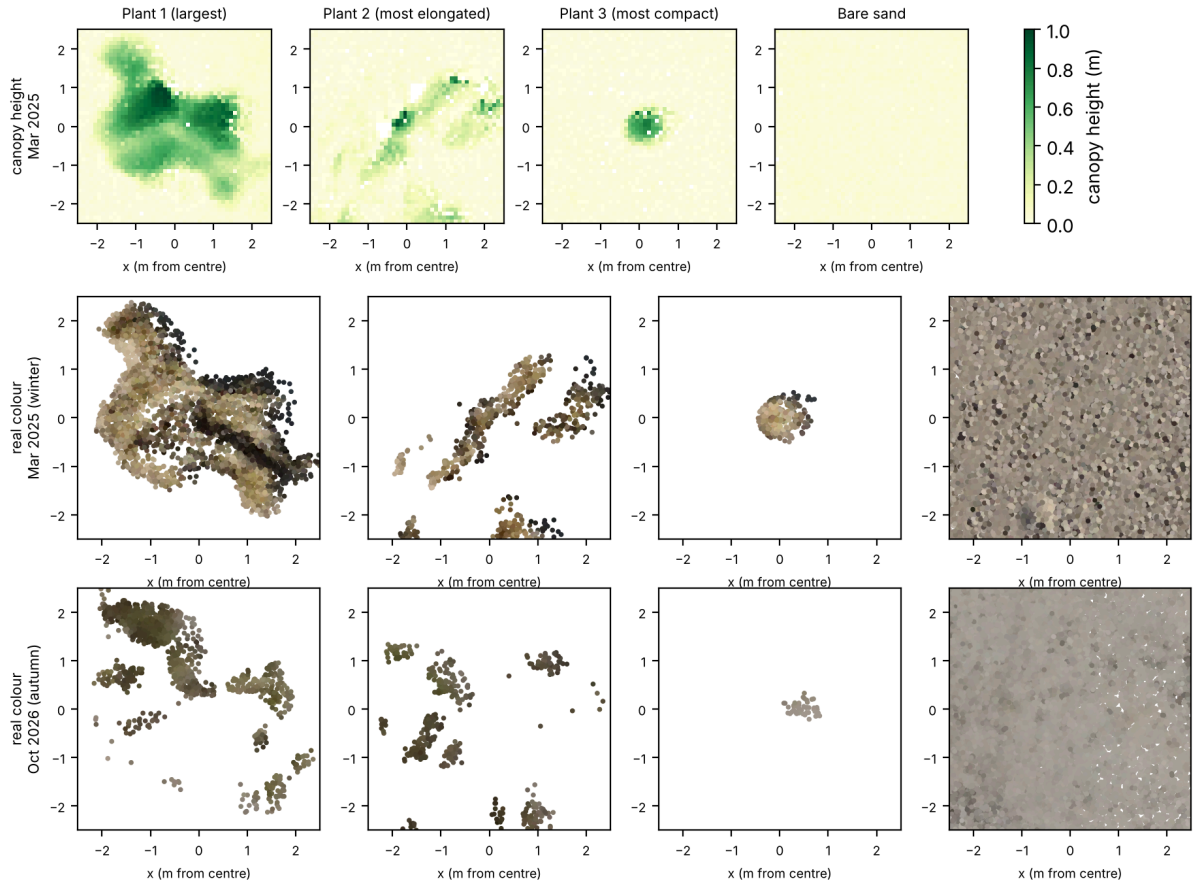


Figure 2: The three example plants with bare sand (March 2025). Top: canopy height; middle: true colour in March 2025 (winter); bottom: true colour in October 2026 (autumn).

Plant	Points	Footprint (m²)	Max height (m)	Volume (m³)	Linearity	Planarity	Sphericity
patch	n_pts	footprint_m2	h_max	vol_column	linearity	planarity	sphericity
Plant 1 (largest)	4896	10.07	1.106	4.635	0.73	0.248	0.022
Plant 2 (most elongated)	440	1.62	1.137	0.574	0.957	0.027	0.016
Plant 3 (most compact)	276	0.8	0.855	0.369	0.247	0.302	0.451

Table 4: Shape of the three example plants (all points, March 2025).

- **Plant 1** (10.1 m², 1.1 m high, 4.6 m³, 4896 points): a large, dense, multi-lobed clump. Size and position (on the hummocks) suggest marram grass (*Ammophila arenaria*), the dune builder, probably several merged tussocks.
- **Plant 2** (1.6 m², 1.1 m, linearity 0.96): a narrow strip along a dune ridge. This looks like sand couch-grass (*Elytrigia juncea*), which spreads along rhizomes, or a strip of marram.
- **Plant 3** (0.8 m², 0.9 m, sphericity 0.45): a small compact tussock, probably a young marram plant.

These species are our interpretation from size, shape and position. The GNSS points have no species recorded, so they are not verified.

Height above ground and 3D shape separate plants from sand best: sand is planar (planarity near 1) with 3 cm noise, while plants are 0.1 - 1.3 m high with low planarity. Between plants, size, elongation (linearity) and compactness (sphericity) differ clearly (Table 4). Colour helps much less. In March 2025 the plants (brightness 0.39 - 0.56) are about as bright as the sand

(0.55), and in October 2026 two plants are darker than the sand (0.31 against 0.61) but the third is the same as the sand (0.58). The grass is dry and brownish, not green, so we do not use colour in the classification.

5.3 One location through time

We follow a 12 × 12 m area around Plant 1 through all eight flights (Figure 3, Table 5; flights at 570 pts/m²).

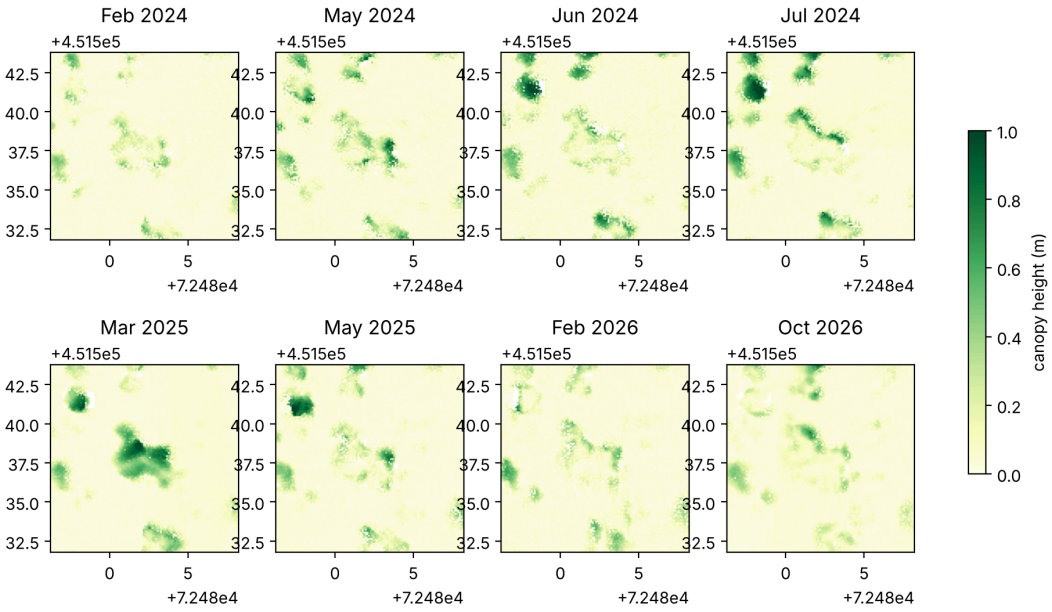


Figure 3: Canopy height in a 12 × 12 m area around Plant 1 (x 72482, y 451538), all eight flights.

Flight	Vegetation cover (m²)	Max canopy height (m)
epoch	cover_m2	max_canopy_height_m
Feb 2024	9.4	0.63
May 2024	16.59	0.99
Jun 2024	17.58	1.11
Jul 2024	18.4	1.2
Mar 2025	19.7	1.11
May 2025	15.08	1.15
Feb 2026	13.29	0.79
Oct 2026	14.85	0.85

Table 5: Cover and maximum canopy height in the area.

In February 2024 the area has little vegetation (9.4 m², highest point 0.63 m). Cover doubles to 18.4 m² and the clump reaches 1.2 m by July 2024. It stays large in March 2025 (19.7 m², 1.11 m), but by Feb 2026 and Oct 2026 the cover is 13 - 15 m² and the highest point 0.8 m. The big clump in the north-west of the area (Mar 2025) is clearly lower in 2026, and the clump in the centre is almost gone in Feb 2026 and only partly back in Oct 2026. Growth in spring and summer and loss in winter are visible, and also a lasting loss at this location. Part of the winter loss is dry leaves lying flat, which are below the 10 cm gate.

5.4 Methodology for separation of the vegetation and sand

Figure 4 shows how the method works around Plant 1, using all points of March 2025. In the 30 × 30 m areas around the three example plants, 7 - 11 % of the points are CSF candidates (Table 6). Most of the candidates that are rejected (4.4 - 6.9 % of all points) are sand ripple and slope breaks. Only 1.2 - 4.0 % of all points end up as vegetation. Figure 5 shows the result on a profile through Plant 1: the plant is vegetation (green), the ground model passes under it, and the sand next to it stays ground.

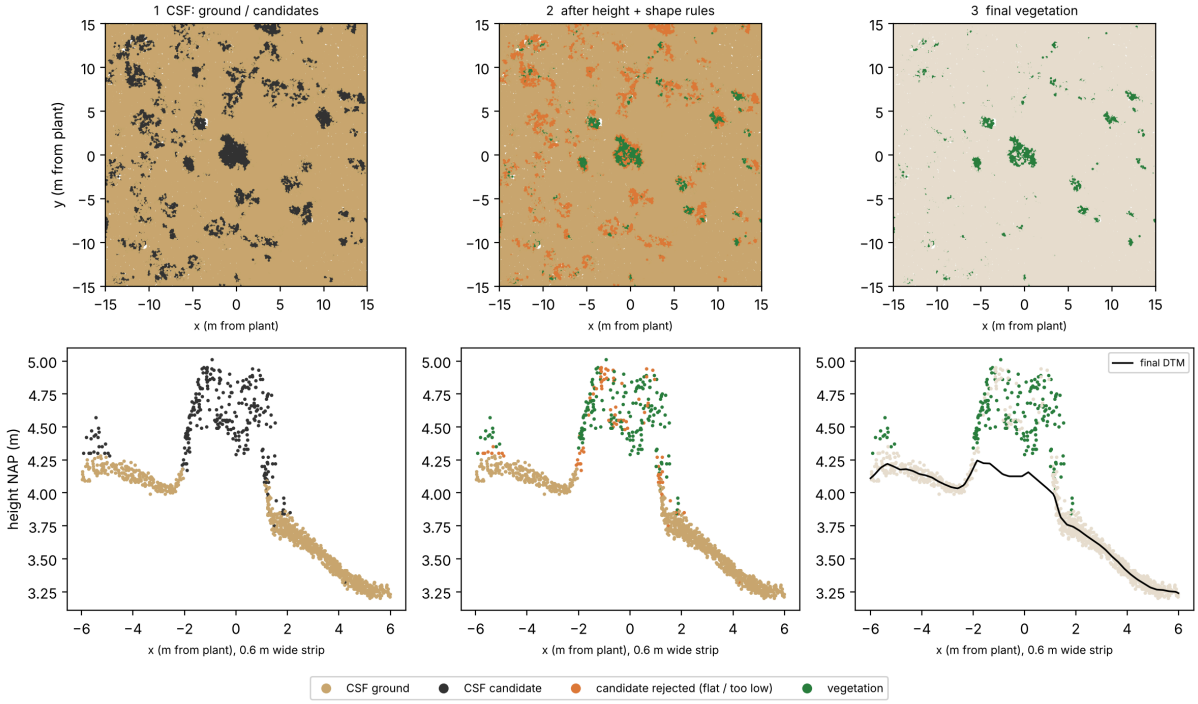


Figure 4: The stages of the separation around Plant 1 (30 × 30 m area, all points). Top: map view; bottom: 0.6 m wide profile. 1: CSF ground and candidates; 2: candidates after the height and shape rules; 3: final vegetation and final ground model.

Area	Points	CSF ground (%)	CSF candidates (%)	Rejected (%)	Vegetation (%)
plant	points	CSF ground %	CSF candidates %	rejected %	vegetation %
Plant 1 (largest)	609884	89.41	10.59	6.61	3.98
Plant 2 (most elongated)	517231	92.62	7.38	4.35	3.03
Plant 3 (most compact)	378069	91.9	8.1	6.94	1.17

Table 6: Share of points at each stage in the 30 × 30 m area around each example plant.

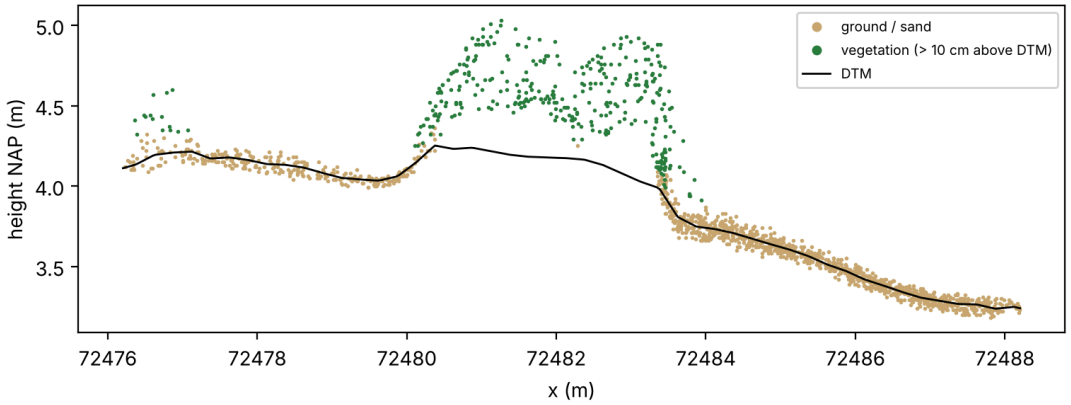


Figure 5: A 0.6 m wide profile through Plant 1: points coloured by class, with the ground model (DTM).

Why not just CSF? Table 7 shows the result of our method and of other settings in one 60 × 60 m test area at full density. The first version of the CSF code (without gates) gave 99 978 "vegetation" points, of which 47 % were less than 10 cm above the ground, i.e. mostly sand. With the height and shape gates none is below 10 cm. CSF settings matter: a finer cloth (0.3 m) hugs the plants and shrinks the largest plant from 4.3 to 1.7 m^3, while a coarser cloth (0.75 - 1.0 m) cuts into the plants and adds 40 - 50 % more volume and points that are probably dune relief.

Variant	Veg. points	< 10 cm (%)	Cover (m²)	Volume (m³)	Largest plant (m³)
variant	W1 vegetation points	W1 points < 10 cm above ground (%)	W1 cover (m2)	W1 tussock volume (m3)	W1 largest plant volume (m3)
report method (CSF res .5, threshold .10, rigidness 2)	51668	0.0	173.9	45.2	4.3
previous rule: all rejected candidates go back to the ground	41310	0.0	159.5	36.1	1.4
first CSF code (res .6, threshold .12, no height / shape gates)	99978	47.1	304.6	39.0	1.7
report rules + CSF res .6, threshold .12	56421	0.0	177.3	47.0	3.9
CSF rigidness 1	60212	0.0	194.4	53.0	4.7
CSF rigidness 3	50613	0.0	170.6	43.9	4.3
CSF threshold .05	56868	0.0	189.7	49.2	4.2
CSF threshold .15	40408	0.0	131.7	35.9	3.9
CSF cloth resolution .3	38562	0.0	146.3	33.8	1.7
CSF cloth resolution .75	76255	0.0	219.7	64.5	4.5
CSF cloth resolution 1.0	81550	0.0	234.7	69.1	4.9
CSF cloth resolution 1.0, threshold .05	87646	0.0	249.2	73.8	4.9
DTM = 25th percentile of ground per cell	65933	0.0	220.0	60.2	4.6

Table 7: Our method (first row) and variants in a 60 × 60 m test area (full density).

The ground under plants. The ground below a plant cannot be seen. Figure 6 shows the ground model we use (black) and two alternatives: a ground model made from the sand around each plant (red) and one with narrow peaks removed (blue). Both alternatives count any mound under a plant as plant, so they are upper bounds. For Plant 1 the volume goes from 4.6 m^3 (used) to 6.4 or 5.6 m^3 (+22 to +38 %). For Plant 3 it hardly changes (0.37 to 0.47 or 0.41 m^3). For Plant 2 it jumps from 0.6 to 2.4 or 2.8 m^3, because the plant sits on a broad dune ridge and the alternatives cut straight across the ridge, so this is an overshoot and not a real value. The ground model we use therefore follows the sand around the plants and rises under them, and if the true ground under dense plants is lower, our volumes and biomass are underestimated by up to a few tens of percent.

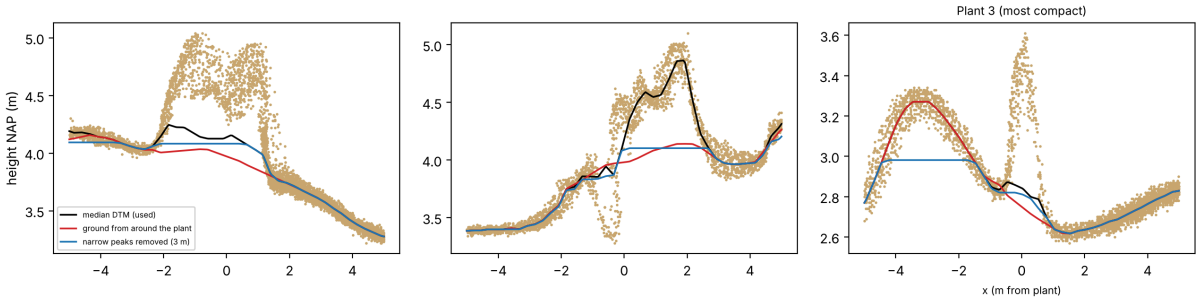


Figure 6: Ground under the three plants: the DTM used (black), a ground model from the ground around the plant (red) and a model with narrow peaks removed (blue). The last two are upper bounds.

5.5 CSF Parameter Tuning

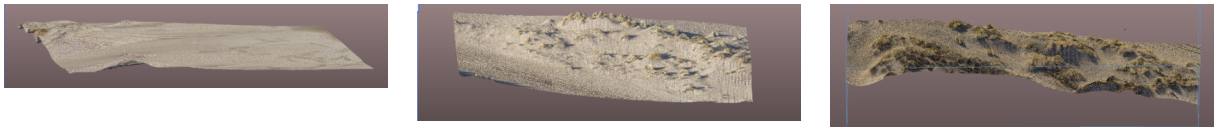


Figure 7: Vegetation types for CSF parameter tuning

resolution = 1.0	rigidity = 2.0	threshold = 0.5

Table 8: Parameter configuration for the highest similarity ratio

Finding the best parameter to represent DTM is done through comparison of three distinct landscape types; bare sand, sparse vegetation, and dense vegetation via cropping of raw dataset (Figure 7). The optimal parameters are selected by identifying the combination that yields the highest overlap in classified data points via the Jaccard similarity as it measures the percentage of shared results of all unique results Table 8. The method is executed not to yield the highest accuracy in terms of ground classification, but to find a combination that achieves the highest degree of similarity across different ground classification combinations.

5.6 Ground-Vegetation separation

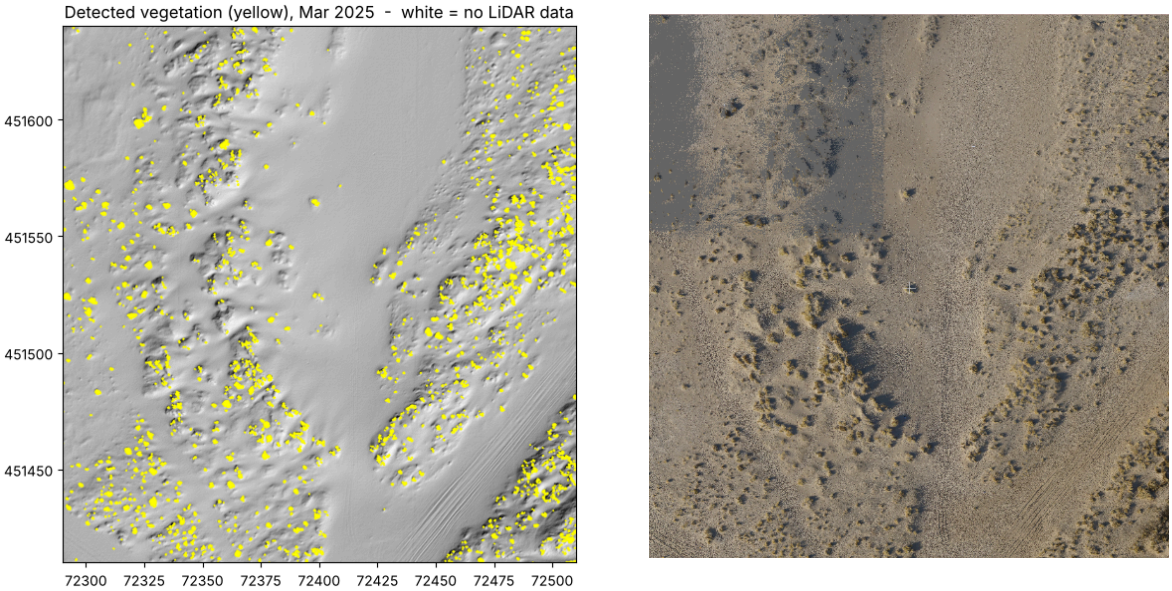


Figure 8: Identified tussocks (left) and RGB point cloud (right) of 2025-03-27 dataset

Figure 8 visualizes the identified vegetation tussocks post-refinement on the left, alongside the RGB point cloud derived from the original, unprocessed point cloud on the right. The identified tussocks point cloud accounted to 0.96% of all points point vegetation and non-vegetation. It is also overlaid on a hillshade map to better illustrate the relationship between DTM slope steepness and tussock distribution. Examining the relationship, most tussocks are identified on the ridge or along the slope of the coastal dune landform. This is consistent with literatures stating the dominance of marram grass found on the area atop of the foredune ridges.

However some obvious misclassification can still be identified when examining the side-by-side figures of the classified tussock and the RGB imagery (Figure 8). Fewer points are classified as vegetation than the RGB imagery initially suggests. However, this observation may be biased as the RGB imagery itself does not necessarily allow for the clear identification of individual tussocks. Moreover, other identified tussocks overestimate the presence of vegetation. Additionally, applying a 10 cm height threshold for vegetation filtering may fail to detect low-lying vegetation, while rough sand or steep slopes can lead to false detections.

5.7 Vegetation characterization method and sensitivity

Once separation between vegetation and ground is distinguished, identified tussocks go through volumetric reconstruction. This design choice is backed by literature findings suggesting higher accuracy yield in this approach, instead of other machine learning methods for shrub or grassland vegetations [8]. Within the volumetric reconstruction, several reconstruction method is also adapted, as different studies suggest different approaches to achieving optimal reconstruction results [8, 9, 11–15, 20, 21], our method aims to evaluate these various method to our dataset. The parameter of each reconstruction method is also evaluated to examine how its change affect reconstruction output. To assess this parameter sensitivity, the tested parameter is isolated and tested across a range of values, while holding all other factors constant.

Voxel size (m)	Volume in m^3
0.05	12.35
0.10	75.98
0.15	178.50
0.20	289.88

Alpha radius (m)	Volume in m^3
0.10	4.80
0.15	29.62
0.20	50.34
0.25	65.00

Table 9: Sensitivity of reconstructed volumes to changes in (a) voxel size and (b) alpha radius.

Larger voxel size results in less spatial resolution of vegetation. Each voxel encompass larger empty space between vegetation points, thus potentially increasing the reconstructed volume.

Both method reveal a similar pattern of sensitivity when a parameter is changed. Increasing the voxel size and alpha radius by 0.05m yielded in a progressively smaller relative increase (or factor multiplication of reconstructed volume) as the parameter value becomes larger. This indicates that both parameters are more sensitive to parameter changes at smaller voxel sizes or alpha radius, yet as parameter value increases, the relative influence on the volume reconstructed gets smaller. The finding highlights the importance of parameter selection as each parameter setting may have substantial influence on the reconstructed volume of identified tussocks.

5.8 Volume to biomass change

Various literatures reported different ways of transforming transforming vegetation characteristic or volume to biomass, depending on type of vegetation. These literatures also suggest the possibility of allometric equation requiring data of diameter breast height (DBH), wood density

and volume as input [22–25]. Several methods were evaluated to assess which method produces biomass estimation that aligns closest with the validated biomass measurements provided by Sander Vos.

$$B_{\text{est}} = aV^b \tag{1}$$

$$B_{\text{est}} \in \left(\frac{7.1}{V}, \frac{10.3}{V}, \frac{13.5}{V}\right) \tag{2}$$

$$B_{\text{est}} = V\varrho \tag{3}$$

B_{est} = biomass estimated in g per m^3

V = volume m^3 from the chosen volume reconstruction method

ϱ = wood density in g per m^3 (Equation 3)

Few attempts are made to compute volume transformation into biomass. The first attempts tries to identify nearby tussocks from the validated xy-coordinate, and fit a power-law equation to find a and b values of the power-law equation (Equation 1). Two tussocks were identified within a 2-meter radius from each xy-coordinate. However, the result yielded a negative b-value, which is counterintuitive as it implies that as volume increases, biomass decreases. This may be attributed to the misidentification of the actual tussock used in the validated biomass estimation.

The second attempt is based on [26] Above Ground Biomass (AGB) allometric equation (Equation 2), as implemented in workflow **B3 - Volume to biomass** . The study provides measured above and below ground biomass values for four common coastal dune grass species which include American Beachgrass (*Ammophila brevihulata*). While the study was conducted in US Atlantic coast and not the Zandmotor, the study still provides useful reference for biomass estimation as *A. brevihulata* provide similar vegetation growth form with Marram grass [27]. The reported above ground biomass of *A. brevihulata* is 10.3 ± 3.2 g, which is used as an initial reference to estimate biomass from the volumetric measurements obtained via the UAV LiDAR point cloud. The result however cannot be validated as the previous power-law equation proves the possibility of misidentifying tussocks during the process. Thus, any subsequent validation would be unreliable as it would not take into account for these possible misidentifications.

Column	Volume	AGB low	AGB mid	AGB high
70	0.213684	0.033227	0.048202	0.063178
218	0.055187	0.128652	0.186637	0.244621
258	0.094891	0.074823	0.108545	0.142268

Column	Volume	AGB low	AGB mid	AGB high
70	0.012750	0.556863	0.807843	1.058824
218	0.005000	1.420000	2.060000	2.700000
258	0.006875	1.032727	1.498182	1.963636

Column	Volume	AGB low	AGB mid	AGB high
70	0.020401	0.348027	0.504884	0.661741
218	0.006558	1.082676	1.570643	2.058609
258	0.005745	1.235784	1.792757	2.349731

Column	Volume	AGB low	AGB mid	AGB high
70	1.654330	0.004292	0.006226	0.008160
218	0.208731	0.034015	0.049346	0.064677
258	0.532594	0.013331	0.019339	0.025348

Table 10: AGB estimates based on (a) column, (b) voxel, (c) alpha-shape, and (d) hull volumes.

The last attempt of volume to biomass conversion (Equation 3) seen on workflow **C1 - Quality Check of the biomass estimate** checks for the March 2025 epoch and processes point cloud dataset that falls within the destructive field biomass sampled and validated by Sander Vos (biomass_measurements.csv). Vegetation points within a 1-meter radius of each

validation point is then isolated. Once isolated, the point cloud is converted from physical dry weights into volumetric densities (kg/m^3) to allow for point of comparison the model's default or custom placeholder to provide direct point of comparison against the model's placeholder density values (ϱ). Across all seven validated vegetation tussocks, the measured densities ranged from a minimum of 0.018 kg/m^3 to a maximum of 8.648kg/m^3. To ensure fair comparison, the dataset was filtered to isolate only the subset of samples where the LiDAR successfuly captures dense vegetation (more than 10 point within 1-meter radius), resulting in 3 out of the 7 validated samples. When compared with the default placeholder density configuration set as (ϱ), the filtered empirical validated points closely aligned with the lower and median bounds.

rho low (kg/m3)	rho mid	rho high	samples min	samples median	samples max
1.50	1.64	3.30	1.50	1.64	8.65

Table 11: Comparison of placeholder (ϱ) parameters with empirical field sample density statistics.

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5.9 Uncertainty

Uncertainty can be separated into two segments. **Structure**, deciding how volume is reconstructed. This also regards point cloud quality, the vegetation-ground separation, and tussock identification. **Conversion** is the process that turns a volume into biomass estimations (kg/m^3).

5.9.1 Structure: point cloud and method

Point cloud density and quality. Table 1 examining the point cloud characteristic, mentioned how single returns only allow UAV LiDAR to see the top of a plant and the sand in between, blocking the view of the ground. This makes the separation between ground and non-ground more challenging. Moreover, 3 cm of noise is also reported on bare sand. When the height gate is also implemented, 10 to 20 cm of dry vegetation that falls below the height gate is not quantified for volume nor biomass.

Tussock parameters. 27 combinations of DBSCAN is ran with epsilon between 0.15 to 0.30, minimum points between 5 to 12, and minimum height between 0.08 to 0.15 meters. The total volume ranges between 77.1 to 156.1 m^3 to assess how the different parameter combinations of the algorithm. The same iteration is also performed for tussock identification, using its different parameter combinations. Once these different combinations are attested, the lowest, original volume and the highest recorded volume are retained. This provides an estimate of the structural uncertainty that comes with the different parameter combinations resulting in very different volumetric reconstructions. This is especially relevant since validation data is little to non-existent.

Ground model. With every height measured from the DTM, the volumetric reconstruction depends highly of how ground parameter is chosen. Changing the cloth resolution from 0.75 to 1.0m changes the volume reconstruction significantly, from 34 to 72% respectively. Measuring the DTM based on its 25th percentile, also changes the volume reconstruction for 155%. These reported findings as observed from **B4 - Uncertainty (Notebook)** is denoted as uncertainty as no point of comparison to validation can verify which ground parameter selection best predicts the best volume reconstructed for the biomass estimation.

Volume method. At the three biomass samples the convex hull volume is more than 100 times the voxel volume (Table 10), and the voxel and alpha volumes grow by a factor 23 and

14 over the tested settings (Table 9). Our biomass is therefore only meaningful together with the column volume.

Tussock identification. Some plants are merged or split, and some objects are misclassified (Figure 8). Tussock counts are therefore less reliable. This is further supported by the failure to identify tussocks used for the data validation. As no corresponding tussocks were identified, the vegetation sampled for validation can be assumed to be misidentified during the vegetation-tussock identification process.

5.9.2 Conversion: volume to biomass

$B_{\text{est}} = V\rho$ with $\varrho = 1.5$ / 1.64 / 3.3 kg/m^3 was deduced, resulting in a range of biomass between 210.6 to 463.2 kg. However, four things can be accounted for to the uncertainty

- **Very few samples.** Only three vegetation samples were identified from the 1 meter distance buffer(1.50, 1.64 and 8.65 kg/m^3). The low and mid ϱ are the minimum and median of these same samples, thus cannot be validated. The highest sample (8.65) is above the "high" value (3.3). With $\varrho = 8.65$ the biomass would be about 1200 kg, so the range above is not a true upper limit.
- **Matching samples to plants.** We matched samples to LiDAR tussocks by position and the matching was uncertain (the power-law fit gave a negative exponent). We also have not checked that the sampled volume in the field is defined like our column volume, but ϱ is only valid if they agree.
- **One date, one density for everything.** The validated samples are from March (dry grass), by one person, without species specification. Using one ϱ for different types of vegetation species sizes, throughout different season, is an ambitious assumption. Literature from S. L. Walker and J. Zinnert [26] also denoted using another family of marram grass, which could not be validated in terms of its exact difference it can account to the biomass prediction

5.9.3 Overall Uncertainty

Uncertainty is present in each step of the methodology. Thinning was initially conducted to improve computational robustness, however, affects the accuracy of the predicted ground, leading to less accurate separation between ground and vegetation, up to volumetric reconstruction. CSF ground filtering and additional conditions application, where small changes in parameter affect the volume reconstruction, thus biomass estimation. One of the most significant uncertainty is however the conversion from volumetric reconstruction to biomass estimation. Literatures review conducted did not reveal one single equation of how volumetric reconstruction directly translate to biomass estimation, in the fixed context of Zandmotor, coastal dune landscape. This uncertainty is further amplified by the lack of validated data against which the reconstructed volumes can be compared. However, methodological and design choices were guided from existing literatures, providing rationale for few approaches to help justify assumptions in the face of encountered uncertainties.

5.10 Quality of the result

Ground height against GNSS. The GNSS height minus the DTM is between 43.25 and 43.47 m for the ten points. This is nearly constant, as it should be, and the median (43.39 m) is the ellipsoid to NAP offset. After removing it the RMSE is 7.6 cm and the largest error is 14.1 cm (Figure 9, Table 12). Errors have both signs, so there is no systematic bias, and they do not grow with plant height (r = 0.28). The three largest errors (points 1, 2 and 9, −11 to

−14 cm) are at plant bases where the sand cannot be seen from above. The error is about the size of the 10 cm height gate, so very low plants (10 - 20 cm) are uncertain, larger plants are not affected.

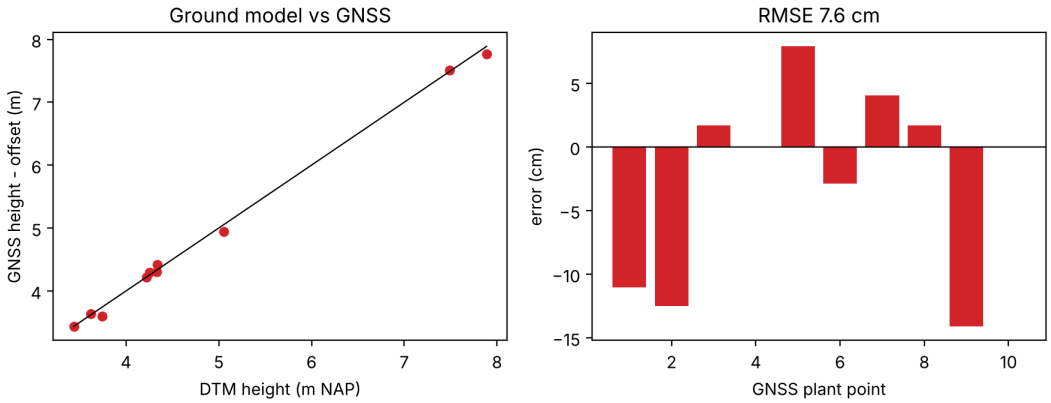


Figure 9: Ground model against the ten GNSS heights (offset removed, left) and error per point (right).

Point	GNSS (m, ellipsoid)	DTM (m NAP)	Plant height (m)	GNSS − DTM (m)	Error (cm)
GNSS plant point	GNSS height (m, ellipsoid)	DTM height (m NAP)	plant height within 0.5 m (m)	GNSS - DTM (m)	minus offset (cm)
1	48.34	5.05	0.42	43.282	-11.1
2	51.16	7.89	0.11	43.268	-12.5
3	50.9	7.49	0.68	43.41	1.7
4	47.61	4.22	0.71	43.393	0.0
5	47.81	4.34	0.71	43.472	7.9
6	47.7	4.33	0.83	43.364	-2.9
7	47.69	4.25	0.75	43.434	4.0
8	47.03	3.62	0.25	43.41	1.7
9	46.99	3.74	0.88	43.252	-14.1
10	46.83	3.44	0.84	43.393	-0.0

Table 12: Ground height check at the ten GNSS points.

Detection at GNSS plant points. For nine of the ten points a vegetation point lies within 0.5 m (median distance 0.04 m). Only point 2 is missed (0.65 m). It lies on the highest ground (7.9 m NAP), a dune crest. For comparison, only 17 % of 5000 random places have vegetation within 0.5 m (median distance 2.0 m), so the match is not a coincidence. All points have 1600 - 3800 LiDAR points within 1 m, so the misses are not data gaps.

Detection at biomass samples. At the three samples with plants the LiDAR finds 164, 296 and 32 vegetation points within 1 m, at the four nearly empty samples it finds none. Presence and absence agree in 7 of 7 samples. This tests detection, not the amount of biomass.

Repeatability. Table 13 gives the noise of the ground model between flights on flat sand. It is small for flights close in time and with similar density (σ = 2 - 4 cm, May - Jul 2024 and Mar - May 2025) and large between seasons and years (σ = 7 - 15 cm). That is not only noise: flat sand also moves a little. The level of detection is therefore different for each pair and we use it in the change maps.

From	To	Median change (m)	σ (m)	LoD 95 % (m)
from	to	median change (m)	sigma (m)	LoD 95% (m)
Feb 2024	May 2024	0.006	0.098	0.192
May 2024	Jun 2024	0.019	0.025	0.049
Jun 2024	Jul 2024	-0.001	0.039	0.076
Jul 2024	Mar 2025	0.052	0.119	0.232
Mar 2025	May 2025	0.0	0.022	0.044
May 2025	Feb 2026	0.03	0.148	0.291
Feb 2026	Oct 2026	-0.014	0.071	0.139

Table 13: Level of detection (LoD) of ground change on flat cells for consecutive flights.

Summary of reliability. Detection agrees with the field data (7 of 7 samples, 9 of 10 GNSS points) and the ground model follows the GNSS heights to 7.6 cm. The weak points are the ground under dense plants, plants below 10 cm, and a small set of reference data (ten GNSS points, three usable samples).

5.11 Spatial distribution

Figure 10 shows vegetation cover per 5 m block for all flights (flights at 570 pts/m²). In every flight the plants sit on the two hummocky belts: the north-east corner ($x > 72470$, $y > 451540$) and the west and south-west side ($x < 72380$). The flat plain in the middle stays almost bare. Cover is weakly related to the terrain (Table 14): it is higher on steeper ground (Spearman $r = 0.21$ - 0.37) and, since 2024, slightly higher on higher ground ($r = 0.01$ to 0.30). This is what we expect: plants establish where there is relief that traps sand, but a correlation of 0.2 - 0.4 means that slope explains only a small part of where plants grow.

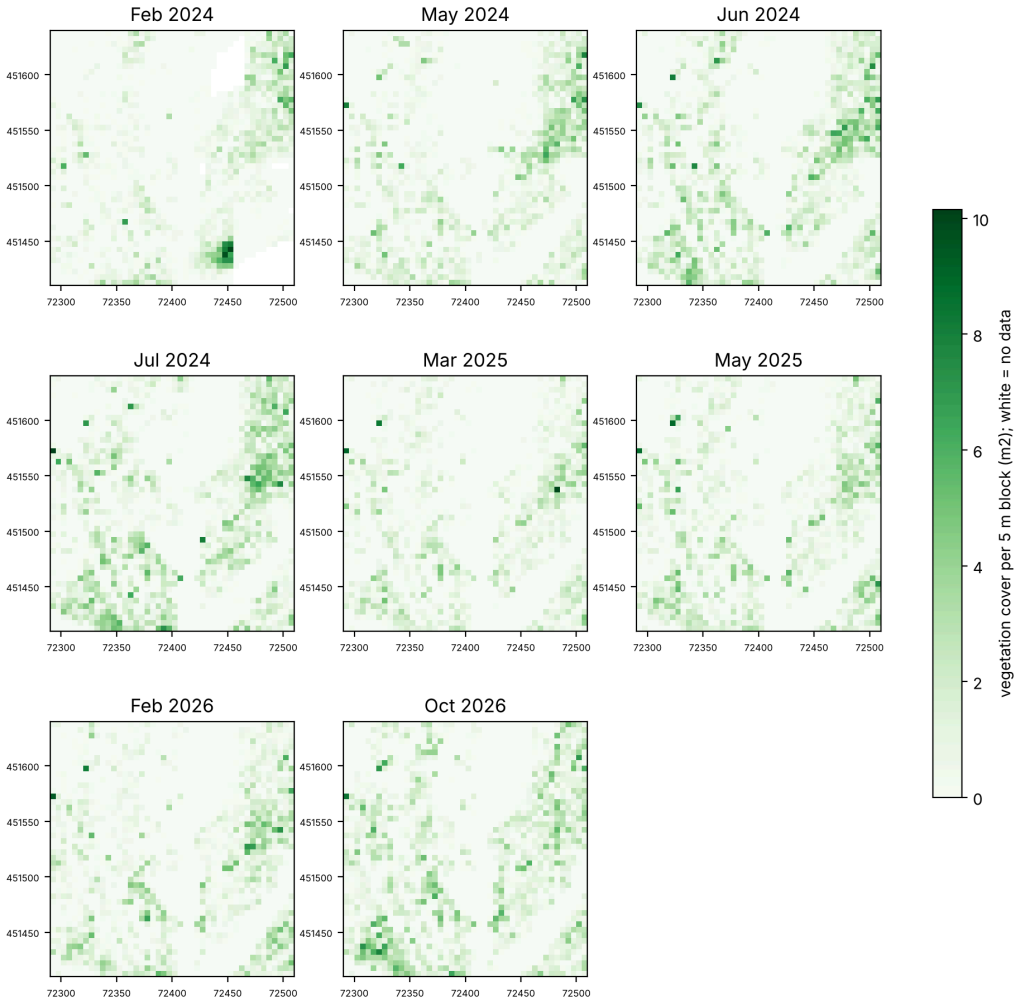


Figure 10: Vegetation cover per 5 m block for all eight flights (white = no data).

Flight	**Terrain variable**	**Spearman r**
epoch	driver	spearman r
Feb 2024	slope_deg	0.21
Feb 2024	z_nap	0.01
May 2024	slope_deg	0.31
May 2024	z_nap	0.11
Jun 2024	slope_deg	0.37
Jun 2024	z_nap	0.23
Jul 2024	slope_deg	0.32
Jul 2024	z_nap	0.18
Mar 2025	slope_deg	0.31
Mar 2025	z_nap	0.25
May 2025	slope_deg	0.32
May 2025	z_nap	0.27
Feb 2026	slope_deg	0.23
Feb 2026	z_nap	0.18
Oct 2026	slope_deg	0.25
Oct 2026	z_nap	0.3

Table 14: Rank correlation between vegetation cover per block and terrain slope and elevation.

5.12 Change in time

On the area covered by all flights (46 575 m²) cover and biomass follow a yearly cycle on top of a net increase (Figure 11, Table 15). From February to July 2024 cover grows from 912 to 1603 m² (x 1.8) and biomass from 310 to 693 kg (x 2.2). By March 2025 it has dropped by about half (869 m², 350 kg), and it grows again to 1138 m² in May 2025. February 2026 (1072 m²) is higher than February 2024 and October 2026 (1569 m², 676 kg) is nearly as high as the July 2024 peak. Over the whole period cover increased by 72 % and biomass by 118 %. Plants (tussock number) rise from 1009 to 1805. The densest flights (2024) were thinned to the same density as the others, so these differences are not caused by density.

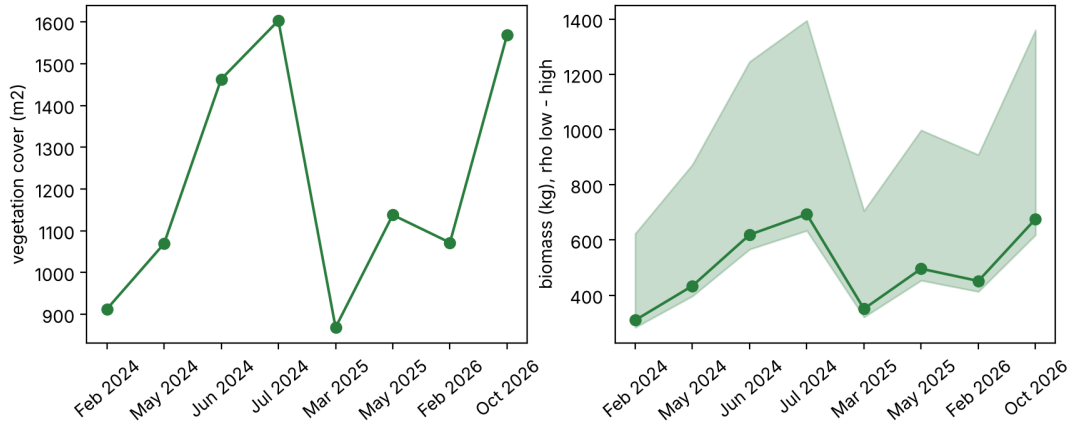


Figure 11: Vegetation cover and biomass of the eight flights (common area 46 575 m², all at 570 pts/m²). The band is ρ low to high.

Flight	Cover (m²)	Tussocks	Volume (m³)	Biomass mid (kg)	low (kg)	high (kg)
epoch	cover_m2	tussocks	volume_m3	biomass_mid_kg	biomass_low_kg	biomass_high_kg
Feb 2024	912.35	1009.0	188.77	309.58	283.15	622.94
May 2024	1068.93	1029.0	264.47	433.73	396.71	872.76
Jun 2024	1462.83	1482.0	377.48	619.06	566.21	1245.67
Jul 2024	1603.02	1602.0	422.62	693.1	633.93	1394.64
Mar 2025	868.58	1085.0	213.71	350.48	320.57	705.24
May 2025	1137.86	1298.0	302.36	495.87	453.54	997.79
Feb 2026	1071.64	1293.0	275.32	451.52	412.98	908.55
Oct 2026	1568.93	1805.0	412.25	676.08	618.37	1360.41

Table 15: Time series of the common area.

Two cautions. First, a higher count of "tussocks" can mean that large plants have split, so cover and biomass are more reliable than counts. Second, the seasonal swing contains true plant growth and dieback and also changes in how well we can see plants (green leaves in summer, dry flat leaves in winter), so the winter minimum should not be read as dead plants only.

Local change. Table 16 gives the largest gains and losses in biomass per 5 m block between Feb 2024 and Oct 2026. The largest gains (+5.7 to +9.2 kg per block) are on the west and south-west side (x 72290 – 72330), where new plants grew on bare sand. The largest losses (−2.7 to −11.3 kg) are in the east and south-east (for example x 72452, y 451442).

x	y	Slope (°)	Biomass Feb 2024 (kg)	Biomass Oct 2026 (kg)	Change (kg)
x	y	slope_deg	biomass_first_kg	biomass_last_kg	change_kg
72452.5	451442.5	3.9	11.34	0.0	-11.34
72492.5	451612.5	8.1	4.22	0.2	-4.02
72502.5	451572.5	4.1	4.45	1.35	-3.1
72357.5	451467.5	5.3	3.18	0.43	-2.75
72502.5	451587.5	8.4	3.13	0.41	-2.72
72327.5	451602.5	9.1	0.33	9.5	9.17
72292.5	451572.5	10.7	0.41	7.34	6.93
72322.5	451597.5	9.6	0.25	6.67	6.42
72327.5	451437.5	6.8	0.0	6.36	6.36
72307.5	451442.5	3.7	0.0	5.71	5.71

Table 16: Largest losses (first five rows) and gains (last five rows) of biomass per 5 m block.

Ground change. Figure 12 shows the change of the ground model, and only where it is larger than the level of detection (0.42 m for Feb 2024 to Mar 2025, 0.43 m for Mar 2025 to Oct 2026 and 0.79 m for the full period). The ground rose on the vegetated hummocks in the north-east and in the west: more than 0.8 m on many of them between Feb 2024 and Oct 2026. Erosion of about 1 m is seen in a few elongated forms around x 72385 - 72410, y 451465 - 451500, where sand is moving. We only show changes larger than the level of detection, so smaller changes may also be real. The ground change on plants includes the growth of the plants themselves, because the ground under a plant is interpolated.

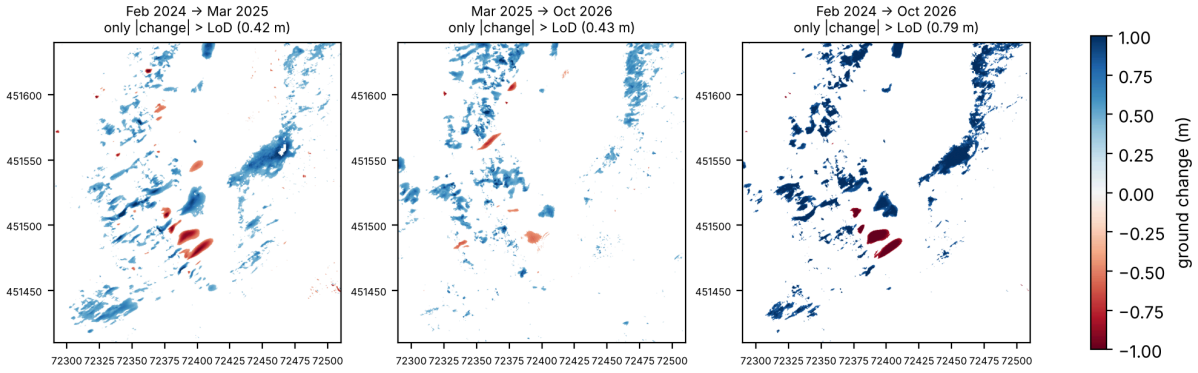


Figure 12: Ground change between flights, only where larger than the level of detection (red = erosion, blue = accretion).

6 Discussion

Limitations.

(1) The ground under dense plants is hidden, so volume and biomass of large plants are probably underestimated by up to a few tens of percent (Section 5.4).

(2) Plants below 10 cm are missed, so cover is a lower limit.

(3) The GNSS check uses only ten points and the geoid offset is estimated from the same points, so it tests the shape of the ground model, not an absolute shift.

(4) The biomass samples are few and were taken on one date in spring (March 2025), we have no samples for summer or other species.

(5) There are no species labels, so growth, loss and biomass cannot be given per species.

(6) Seasonal changes in the point cloud (leaf state, moisture) are mixed with growth.

(7) The CSF settings change the result by tens of percent (Table 7).

(8) Strict conditions applied for vegetation and ground separation can lead to underestimation or overestimation of identified tussocks. This leads to unusable validation data due to inability to identify tussock presence.

(9) Simplified biomass conversion that relies on the assumption of similar vegetation growth between American beachgrass & Marram grass and the nature of the study, in which biomass estimation is very dependent on specific vegetation type, structure and environment.

Recommendations.

(1) Collection of more biomass samples, over species, sizes and seasons, specific vegetation characteristic such as diameter, with a known sampled volume for validation purposes.

(2) Record more GNSS heights, on bare sand as well as at plants, and record the species at every point.

(3a) Fly at the same density and the same time of year, so that changes can be compared without thinning.

(3b) Instead of random thinning, the point clouds could be thinned on a voxel grid (one point per cell), so that all flights get an even point spacing and the same density everywhere. This should make the comparison between flights more consistent, and it could be tested whether it also improves the separation of vegetation and ground.

(4) Add multi-return LiDAR or multispectral data to look into the canopy and to separate species.

(5) Improvement on individual tussock identification, that may improve reliability if identified vegetation across the landscape for better volumetric prediction.

7 Conclusion

UAV LiDAR point cloud were processed through a workflow that combines several algorithms spanning Cloth-Simulation-Filtering algorithm, height-based filtering, DBSCAN clustering to distinguish between ground and vegetation. The reconstructed DTM achieved an RMSE of 7.6cm against GNSS field measurements, indicating a consistent representation of the underlying terrain or ground separation. Moreover, most tussocks identified were located in steeper or

frontal face of the dune, while most bare areas remain unvegetated. Between February 2024 to October 2026, vegetation cover is increased by 72%. Despite seasonal fluctuations observed, overall trend indicates an increase in vegetation. Uncertainty is undeniably encountered during methodology process; this include the sensitivity of parameter tuning to volume reconstruction, vegetation and ground separation, resulting in very different tussock identification and volumetric reconstruction. When different combinatorials were tested over a specified range, volumetric structure has uncertainty as low as 70.43 m^3 and as high as 144 m^3, while default configuration results in 140.66 m^3. Moreover conversion from volumetric structure to biomass, outputs a low threshold of 210 kg/m^3 and high threshold of 464.18 kg/m^3, with default configuration of 230.68 kg/m^3. Thus, both volumetric reconstruction and biomass conversion introduce substantial uncertainty. Additionally, density parameter acting as placeholders to produce biomass range should be interpreted as a scenario-based estimate, rather than a statistically validated confidence interval. Future studies should incorporate more comprehensive field measurements such as diameter at breast height (DBH), stem width, and more validation points. Collecting these additional measurements would provide better reference data for vegetation characteristics and biomass estimation, thus improving the accuracy and study reliability.

8 Work distribution and use of AI

Throughout the development of this assignment, we have been assisted by OpenAi's ChatGPT version GPT-5.6 Luna, Google's Gemini 3 Flash (Free tier) and the free version of Claude AI. We have reviewed and understood all suggestions made by those models, and take full responsibility for the content being submitted.

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